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First and foremost, we express our deep gratitude to God Almighty . We Also thank our families who supported us and our teachers who guides us . THANK YOU ALL

Abstract

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List of Acronyms

MLIR	Multi-Level Intermediate Representation
RL	Reinforcement Learning
Deep RL	Deep Reinforcement Learning
PPO	Proximal Policy Optimization
LSTM	Long Short-Term Memory
AST	Abstract Syntax Tree
IR	Intermediate Representation
LLVM	Low Level Virtual Machine
SIMD	Single Instruction Multiple Data
SCF	Structured Control Flow
SSA	Static Single Assignment
CFG	Control-Flow Graph
MDP	Markov Decision Process
DQN	Deep Q-Network
GAE	Generalized Advantage Estimation
SGD	Stochastic Gradient Descent
RNN	Recurrent Neural Network
TD	Temporal Difference
JIT	Just-In-Time
ReLU	Rectified Linear Unit
ELU	Exponential Linear Unit
AVX	Advanced Vector Extensions
ILP	Instruction-Level Parallelism

Chapter 1

Automatic Assessment of Qur’anic Recitation: Background and State of the Art

1.1 Introduction

The Qur’an occupies a central place in Islamic worship, learning, and daily practice. Its recitation is not treated as ordinary reading: it is a transmitted oral practice in which pronunciation, rhythm, pausing, elongation, and articulation must follow recognised rules. This discipline is known as Tajweed. For learners, Tajweed is often one of the most challenging aspects of Qur’anic study because errors are primarily acoustic: a mistake may be entirely invisible in the written word yet immediately audible in how a sound is produced.

The first part establishes the linguistic and pedagogical background required to understand automatic Tajweed assessment. It begins with an overview of Qur’anic recitation and the system of canonical transmission traditions known as *Qirā’āt* and *Riwayāt*. It then presents the main Tajweed rule families and their acoustic relevance, and discusses the types and severity of recitation errors, because not all mistakes carry the same weight or require the same form of analysis. The chapter also examines the pedagogical role of traditional teacher correction and of the coloured *Mushaf*, a widely used visual learning aid, as these two supports define what an automated system must eventually replace or complement.

The second part of the chapter surveys the state of the art in automatic Tajweed assessment. It surveys the main categories of resources available for automatic Tajweed assessment, reviews the principal research directions and their representative systems, and compares the modelling approaches found in the literature. The objective is not merely to describe existing work, but to identify the gap that this thesis addresses. As the review will show, current systems provide valuable but partial solutions: they perform speech recognition, classify specific rules, or detect phoneme-level errors. However, a complete automatic Tajweed assessment system must achieve more: it must verify the recited content, locate each deviation, identify the applicable rule, distinguish the type of mistake, and produce feedback that is actionable for the learner. No existing system achieves all of these simultaneously.

1.2 Qur'anic Recitation, Tajweed, and Learning Support

1.2.1 Qur'anic Recitation and Oral Transmission

The Qur'an is the sacred text of Islam and its recitation occupies a central place in Islamic worship, education, and oral tradition. Unlike ordinary reading, Qur'anic recitation is governed by a precise set of phonetic, rhythmic, and articulatory constraints. The learner is expected not only to pronounce the Arabic text correctly, but also to preserve the manner of recitation transmitted through recognised chains of oral instruction [13, 14].

This oral dimension is essential for the present work. Many recitation errors cannot be fully understood from the written text alone. Two written sequences may differ in how they are realised acoustically depending on elongation, nasalisation, assimilation, concealment, or pausing. Automatic Tajweed assessment therefore requires a combination of textual reference, acoustic analysis, and rule-aware interpretation none of these three is sufficient on its own.

1.2.2 Qirā'āt, Riwayāt, and Recitation Variants

The Qur'an has been transmitted through recognised canonical readings known as *Qirā'āt* (القراءات), each associated with a well-known imam of recitation and transmitted through specific narrations known as *Riwayāt* (الروايات). The most widely known examples include Ḥafṣ from 'Āṣim (حفص عن عاصم) and Warsh from Nāfi' (ورث عن نافع), both of which differ in Hamzah treatment, vowel realisation, and Madd durations at specific positions. Academic studies on the history of Qur'anic transmission show that these traditions form a structured and documented science of variant reading [18]. The distinctions between transmissions in vowel realisation, Hamzah treatment, and orthographic conventions are well-established in the classical Qirā'āt literature and have been studied extensively by modern scholars [14, 18].

For automatic assessment, this has a direct practical consequence: the expected pronunciation at any given position depends on which transmission the learner follows. A reference-aware system must therefore know not only the canonical text but also the applicable transmission-specific realisations. In this thesis, the focus is placed on Tajweed rules that are common and pedagogically central across traditions Madd, Ghunnah, Ikhfā', Idghām, Iqlāb, and Qalqalah. ~~we can delete this part~~ The system architecture is nevertheless designed to be modular: extending it to a specific Qirā'ah would require only replacing the rule-annotated reference at the alignment stage with one that encodes the realisations of the chosen transmission; all other components remain unchanged.

1.2.3 Fundamental Tajweed Rules

Tajweed (التجويد) is the discipline that regulates the correct pronunciation and performance of Qur'anic recitation. Literally meaning “to do well” or “to make excellent,” it concerns the precise manner in which each letter is articulated, the attributes that distinguish letters from one another, and the rules that govern how sounds interact in context. Its foundations were codified by classical scholars

in authoritative texts that remain the primary references for Tajweed teaching to this day: the *al-Muqaddimah al-Jazariyyah* (المقدمة الجزرية) and *al-Tamhīd fī 'Ilm al-Tajweed* (التمهيد في علم التجويد) of Ibn al-Jazarī [13, 14], the *al-Ri'āyah li-Tajweed al-Qirā'ah* (الرعاية لتجويد القراءة) of Makkī ibn Abī Ṭālib al-Qaysī [5], and the *al-Taḥdīd fī al-Itqān wal-Tajweed* (التحديد في الإتقان والتجويد) of Abū 'Amr al-Dānī [3].

Table 1.1 presents the main Tajweed rule families, their representative examples, their acoustic relevance for assessment, and the category of acoustic analysis they require. This last column is of particular importance: it shows that Tajweed errors are not acoustically uniform, and that different rule families require fundamentally different forms of detection.

Table 1.1: Main Tajweed rule families with Arabic names, acoustic relevance, and required analysis category.

Rule family	Examples	Why it matters for assessment	Analysis category
Makhārij al-Ḥurūf مخارج الحروف (articulation points)	Articulation from the throat, tongue, lips, and nasal passage	Detecting letter substitutions and place-of-articulation errors	Content / articulatory
Sifāt al-Ḥurūf صفات الحروف (letter attributes)	Hams الهمس, Jahr الشدة, الجهر, Shiddah, Rikhāwah الرخاوة, Tafkhīm التفخيم, Tarqīq الترقيق	Detecting acoustic qualities such as emphasis, breathiness, and friction	Local spectral
Nūn Sākinah and Tanwīn النون الساكنة والتنوين	Izhār الإظهار, Idghām الإدغام, Iqlāb الإقلاب, Ikhfā' الإخفاء	Context-dependent nasalisation and phoneme transitions	Transition-oriented
Mīm Sākinah الميم الساكنة	Idghām Shafawī الإدغام الشفوي, Ikhfā' Shafawī الإخفاء الشفوي, Izhār Shafawī الإظهار الشفوي	Lip-related assimilation and concealment at word boundaries	Transition-oriented
Madd المد (vowel elongation)	Madd Ṭabī'ī المد الطبيعي, Madd Mut-taṣil المتصل, Madd Munfaṣil المنفصل, Madd Lāzim المذ لازم	Sustained duration of a vowel over a required number of counts	Duration-oriented
Ghunnah الغنة (nasalisation)	Ghunnah in Nūn التنون, Mushaddadah المشددة, Mīm الميم, Mushaddadah المشددة	Presence and duration of nasal resonance	Duration-oriented
Qalqalah القلقلة (echoing release)	Qalqalah Ṣuġhrā and القلقة الصغرى, Qalqalah Kubrā القلقة الكبرى	Short localised release or echo-like acoustic event at a stop consonant	Burst-oriented
Waqf and Ibtidā' الوقف والابتداء (pausing and resuming)	Waqf Jā'iz الوقف الجائز, Waqf Mamnū' الوقف الممنوع	Phrase boundaries and valid starting positions	Contextual / structural

can be deleted The four analysis categories reflect the distinct types of acoustic evidence needed to evaluate each rule family and serve as organising principles throughout this thesis.

These distinctions motivate the modular design adopted in this thesis: no single model handles all rule families equally well, so each category is matched to an analysis module suited to its dominant evidence type.

1.2.4 Types and Severity of Recitation Errors

Errors in Qur'anic recitation differ in both nature and severity. The foundational Tajweed literature [14, 5, 3] draws a canonical distinction between two categories: *Lahn Jalī* (الحن الجلي), the obvious or manifest error, and *Lahn Khafī* (الحن الخفي), the hidden or subtle error. This distinction is not merely descriptive; it carries direct consequences for how an automated system must be structured.

Lahn Jalī (Content-level errors): It denotes an obvious error that alters the canonical text itself. It encompasses all content-level deviations: letter substitution (replacing one phoneme with another, such as *س* for *ص*), deletion (dropping a letter or syllable entirely), insertion (adding a sound that is not in the text), and vowel distortion (changing a short or long vowel in a way that affects grammatical meaning or word identity). The classical scholars regarded this as the more severe category, since it alters *what* is recited and may change the meaning of the divine text [14]. From a system-design perspective, these errors are detected at the content level, before any rule-level analysis is applied.

Lahn Khafī (Rule-level errors): It denotes a subtler error that concerns *how* the correct content was recited. The recited word is identifiable, but a Tajweed rule has been violated in its delivery: a Madd is shortened below its required duration, a Ghunnah is weakened or absent, Ikhfā' is not realised, or Qalqalah is omitted at a stop. This category is considered less severe than Lahn Jalī because it does not alter the recited text, but it is nonetheless blameworthy for any reciter who knows the rules, and its correction is the primary concern of Tajweed pedagogy [5, 3].

This two-level structure has a direct consequence for system design. A system that operates only at the content level verifying that the correct words were recited will miss the entire Lahn Khafī class. Conversely, a system that evaluates rule execution without first confirming content correctness risks producing rule judgements on wrongly recited segments. A complete assessment pipeline must therefore handle both levels in the correct order: content verification first, rule analysis second.

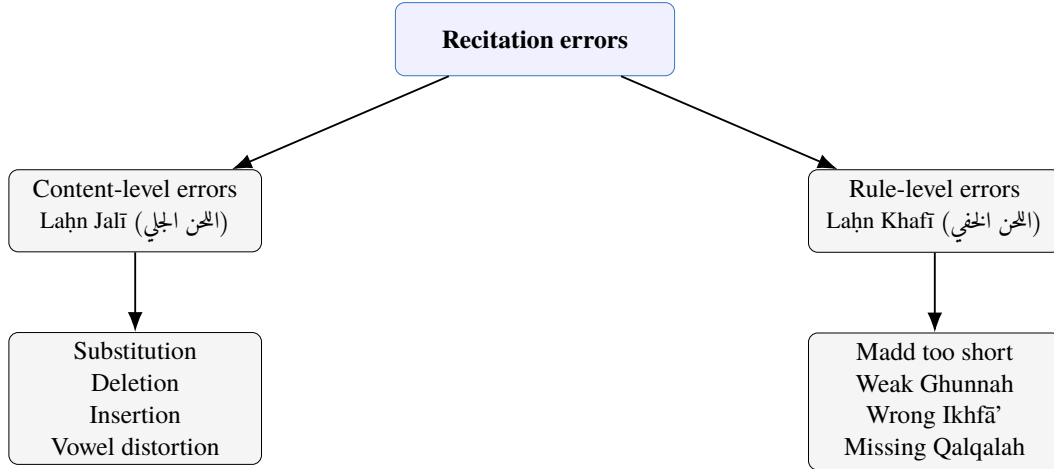


Figure 1.1: Taxonomy of recitation errors: content-level deviations (Laḥn Jalī) versus rule-level mis-applications (Laḥn Khafī), following the classical distinction in Ibn al Jazari [14], al Qaysi [5], al Dani [3].

1.2.5 Traditional Qur’anic Learning

Traditional Qur’anic learning relies on listening, repetition, memorisation, and correction by a qualified teacher [14]. The learner listens to a model recitation, repeats it in small segments, receives correction, and revises the passage until the recitation is stable.

The diagnostic value of the teacher lies not only in detecting that an error exists, but in performing three distinct functions simultaneously: identifying *where* in the verse the error occurred, determining *what type* of error it is whether it concerns the content, a duration rule, a transition rule, or another Tajweed category and explaining *how* it should be corrected. These three functions localisation, classification, and corrective guidance define the minimum specification of an automated assessment system that aims to replace or complement the teacher’s role. A system that only signals the presence of an error without locating it or naming its type provides far less pedagogical value than a qualified human instructor.

1.2.6 The Coloured Mushaf as a Visual Learning Aid

The coloured Mushaf, or *Mushaf al-Tajweed al-Mulawwan* (مصحف التجويد الملون), is a widely used printed learning aid in which colours mark the positions where specific Tajweed rules should be applied. The colour does not alter the Qur’anic text; it encodes the rule attached to a letter or word and makes it immediately visible to the reader.

Colour conventions vary between publishers, but the general principle is consistent: one colour may indicate Madd, another Ghunnah or Idghām, another Qalqalah, and another letters that are written but not pronounced. This visual support is particularly valuable for beginners because many rules depend on context and are not derivable from the plain text alone. The coloured Mushaf therefore provides a rule-aware layer on top of the canonical text, guiding the learner from the written position to the

expected recitation behaviour. Published editions such as the Dar al-Ma'rifah *Mushaf al-Tajweed al-Mulawwan* [9] represent this pedagogical tradition.



Figure 1.2: Example of a coloured Mushaf (مصحف التجويد الملون) used as a visual aid for Tajweed learning. Colours indicate the applicable Tajweed rule at each position.

The conceptual importance of the coloured Mushaf for this thesis extends beyond pedagogy. It demonstrates that effective Tajweed learning depends on explicitly linking three elements: the textual position in the canonical recitation, the applicable Tajweed rule at that position, and the expected acoustic realisation.

1.3 Resources, Related Work, and Research Gap

1.3.1 Speech and Language Resources for Qur'anic Recitation

Developing an automatic Tajweed assessment system requires several distinct categories of resources. As illustrated in Figure 1.3, audio corpora, canonical text resources, rule annotations, and error-labelled datasets each play a distinct and non-substitutable role.

Expert recitation audio: A speech corpus is a principled, structured collection of audio recordings assembled for the purpose of training or evaluating automatic speech processing systems [17]. In the context of Qur'anic recitation, recordings from expert reciters provide clean examples of correct pronunciation, rhythm, and recitation style, and support acoustic modelling or reference-based analysis. They are not, however, sufficient for error detection in isolation, since they contain correct recitation only and carry no information about how errors manifest acoustically.

Textual and linguistic resources. Textual resources such as the Qur'anic Arabic Corpus [10] provide linguistic information including word-level morphological structure, part-of-speech tagging, and syntactic annotation. For Tajweed assessment specifically, the most important textual resource is not the plain canonical text but a rule-annotated reference that indicates where Tajweed rules apply and which rule governs each position. Such a resource is what allows the system to move from generic pronunciation scoring to rule-specific diagnosis.

Benchmarks and shared evaluation resources. A significant recent development is the emergence of standardised benchmarks for Qur'anic pronunciation assessment. El Kheir et al. [11] introduced QuranMB.v1, the first unified benchmark for Qur'anic mispronunciation detection, together with self-supervised baseline models evaluated under shared conditions. The IqraEval shared task provided a further evaluation framework for Arabic and Qur'anic phoneme-level pronunciation assessment; Ibrahim [15] introduced Hafs2Vec, a system targeting this task at the phoneme level and reporting competitive performance under the shared evaluation protocol. The Tarteel dataset, introduced by Khan et al. [16], is a large crowd-sourced Qur'anic recitation corpus containing over 25,000 audio clips and 67 hours of audio contributed by more than 1,200 speakers, which has supported downstream research on Qur'anic speech technology. These resources collectively indicate that the field is moving toward standardised, reproducible evaluation rather than isolated single-task experiments.

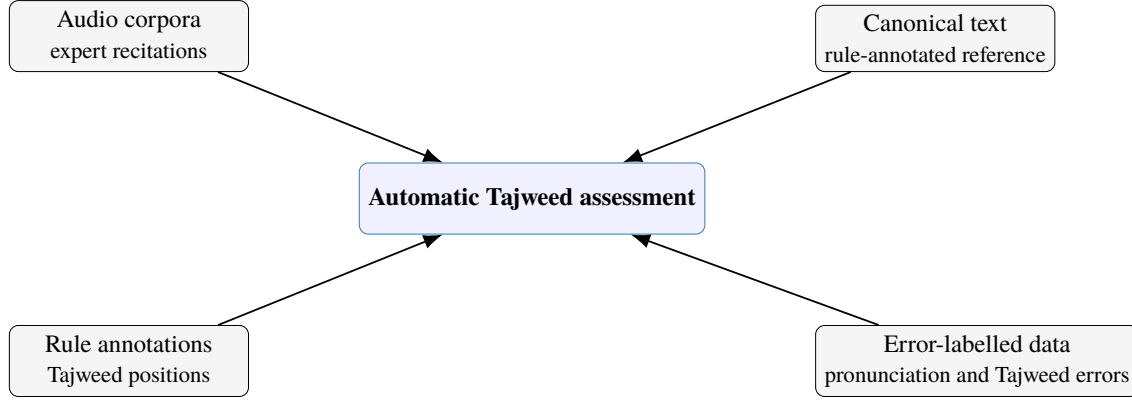


Figure 1.3: Main resource types required for automatic Tajweed assessment.

1.3.2 General Landscape of Related Work

Existing work relevant to automatic Qur'anic recitation assessment can be grouped into four broad directions. Some studies focus on Qur'anic speech recognition or verse identification. Others target the detection of specific Tajweed rules through acoustic classification. A third direction comes from general Arabic mispronunciation detection, where the task is to identify phoneme-level pronunciation errors in Arabic speech by non-native or learner speakers. More recent work explores self-supervised speech models, multimodal fusion architectures, and feedback generation.

Each direction addresses only part of the complete assessment problem. Speech recognition establishes what was recited. Tajweed classification evaluates rule-level correctness for specific rules. Arabic mispronunciation detection provides methods for phoneme-level error identification. Feedback-oriented systems show how detection output can be turned into learner guidance. A complete Tajweed assessment system must integrate all of these directions rather than rely on any single one.

1.3.3 Existing Approaches to Qur'anic Recitation Assessment

Early rule-detection systems. The earliest attempts at automatic Tajweed assessment focused on recognising whether a specific rule had been applied correctly, treating the problem as a classification task. Al-Ayyoub et al. [2] built one of the first comprehensive systems of this kind, targeting eight Tajweed rules across the entire Qur'an. Their system extracted acoustic features from the audio signal and used a machine-learning classifier to decide whether each rule was applied correctly or not, achieving strong accuracy on their test recordings. The work demonstrated that automatic rule detection was feasible but left two important limitations unaddressed: it relied on data that was not publicly available, and it produced only a binary verdict with no explanation of what went wrong or how to correct it.

Sequence-based approaches for temporal rules. A natural next step was to use models better suited to the temporal nature of Tajweed rules. Rules such as Madd and Ghunnah are not about a single moment in the audio they require a sound to be sustained for a specific duration and this makes them

particularly well suited to sequence-based modelling. Al Harere and Al Jallad [4] trained models of this kind for three rules Madd, Ghunnah, and Ikhfa' and showed that this approach outperformed earlier classification methods by a substantial margin. Ahmed et al. [1] applied a similar approach to Arabic mispronunciation detection more broadly, demonstrating that taking into account both the preceding and following phonetic context leads to better error identification. The limitation shared by both works is that they cover only a small number of rules and do not provide any corrective feedback to the learner.

Pattern-based approaches for local events. Some Tajweed rules produce a distinctive acoustic event that is concentrated at a single point in the signal rather than spread over time Qalqalah is a clear example. For these rules, approaches that detect local patterns in the audio are well suited. Salim et al. [19] explored this direction, showing that such methods work well for compact, localised rule events but are less effective for rules that require tracking a signal over a longer stretch of time.

Combining multiple acoustic representations. Another line of research investigates whether combining different ways of representing the audio signal can improve detection performance. Hassan et al. [12] proposed a system that runs two parallel analysis branches one capturing fine-grained local details and one capturing broader acoustic patterns and then merges their outputs before making a decision. Applied to the task of identifying Qur'anic reciters, the combined system outperformed either branch used alone, demonstrating the value of complementary representations. Although reciter identification is not the same as error detection, the principle carries over: different rule families leave different kinds of acoustic evidence, and a single representation may not capture all of them equally well.

Large pre-trained speech models. The most recent direction in the field makes use of large speech models that have been pre-trained on vast amounts of audio data before being adapted to a specific task. Alrashoudi et al. [6] showed that adapting models of this kind to Arabic pronunciation assessment yields excellent results, even outperforming human annotators on certain error categories. Calik et al. [8] confirmed this finding using a different model architecture, also on Arabic speech data. These results establish pre-trained speech models as the current state of the art for recognising what was said and identifying phoneme-level errors though neither study addresses Tajweed rules specifically.

Deployed applications and the feedback gap. Beyond academic research, several applications have been built to support Qur'anic learning in practice. Tarteel, for example, offers recitation verification and memorisation tracking and is widely used. A survey of such tools by Birgun [7] reveals a consistent pattern: these applications are generally good at detecting that an error occurred, but they tell the learner very little about what type of error it was or how to fix it. This gap between detection and explanation has also been noted in the wider mispronunciation research community. Xie et al. [20] showed that pairing a detection system with a language model capable of generating corrective explanations can produce feedback that learners find useful, though their work addresses English and Chinese rather than Arabic or Tajweed.

1.3.4 Evolution of Approaches

Figure 1.4 places the reviewed work in chronological perspective. The field has moved through three broad phases: an early phase relying on hand-crafted audio features and conventional classifiers (2018–2020), a middle phase in which sequence-based models became the standard approach for temporal rule detection (2021–2023), and a current phase in which large pre-trained speech models define the state of the art for content recognition and phoneme-level assessment (2024–present).

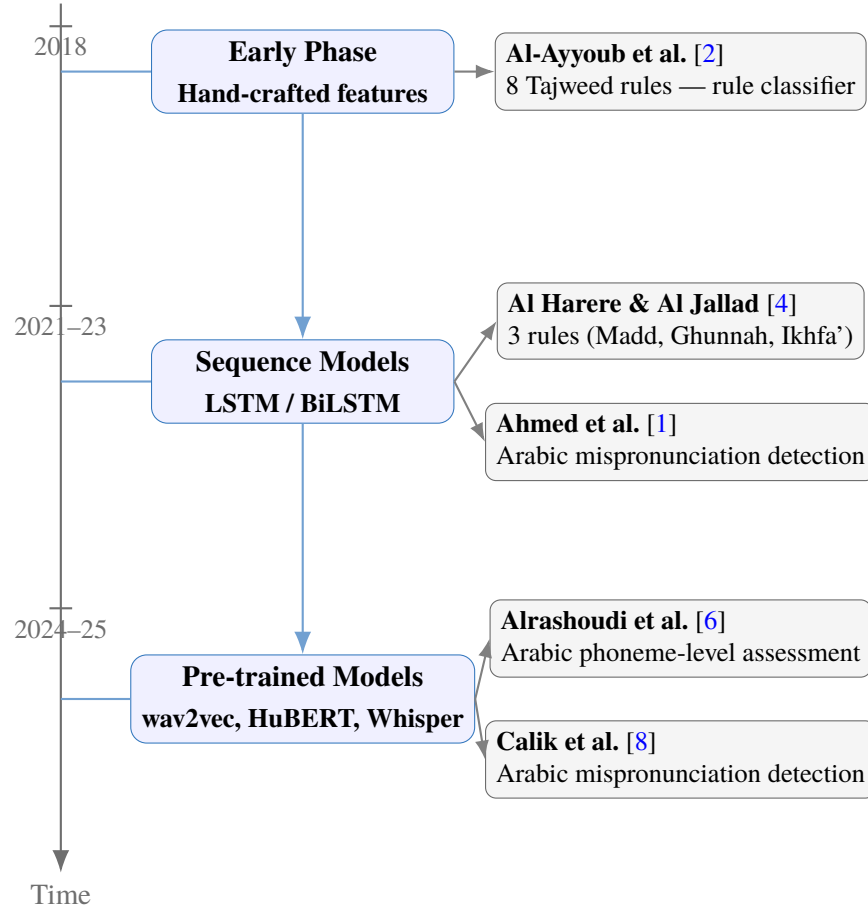


Figure 1.4: Evolution of modelling approaches in Qur'anic recitation assessment and Arabic mispronunciation detection, from early hand-crafted feature systems through sequence-based models to the current generation of large pre-trained speech models.

1.3.5 Comparative Summary of Reviewed Systems

Table 1.2 synthesises the main characteristics of the systems reviewed above across a set of criteria directly relevant to the design of a complete automatic Tajweed assessment system.

Table 1.2: Comparative summary of reviewed systems for Qur'anic recitation assessment and Arabic mispronunciation detection.

Work	Year	Approach	Task / Rules	Dataset	Feedback	Key limitation
Al-Ayyoub et al. [2]	2018	MFCC + CDBN + SVM	8 rules, 16 classes	In-house (3,071)	None	Private data; no feedback
Al Harere & Al Jallad [4]	2023	MFCC + LSTM	3 rules (binary)	QDAT (1,500+)	None	3 rules only; single verse
Ahmed et al. [1]	2023	MFCC + BiLSTM	Arabic MDD (phoneme)	In-house	None	Not Tajweed-specific
Salim et al. [19]	2025	MFCC + CNN	Tajweed law recognition	In-house	None	Local patterns only; limited rule set
Hassan et al. [12]	2025	MFCC + CNN / ResNet50 fusion	Reciter classification	7,144 clips	None	Task mismatch (not error detection)
Alrashoudi et al. [6]	2025	wav2vec / HuBERT / Whisper fine-tuning	Arabic MDD (phoneme)	L2-KSU	None	Not Tajweed-specific; no rule routing
Çalık et al. [8]	2024	Transformer (audio-oriented)	Arabic phoneme MDD	Arabic speaker data	None	Not Tajweed-specific

1.3.6 Comparison of Model Architecture Options

Table 1.3 summarises the main model architectures found in the literature, their strengths and weaknesses, and the evidence supporting each.

Table 1.3: Comparison of model architectures discussed in the reviewed literature.

Model	Strengths	Weaknesses	Evidence from the literature
CNN 1D/2D	Efficient for local acoustic and time-frequency patterns; suitable for short events.	Limited temporal range unless combined with a sequential layer.	MFCC–CNN used for Tajweed law recognition; performance degrades on acoustically similar classes [19].
LSTM / BiLSTM	Designed for sequence data, variable-length input, and temporal dependencies; well suited to duration and transition rules.	Slower to train; may struggle with very long sequences.	MFCC–LSTM achieves 95–97% F1 on three Tajweed rules [4]; BiLSTM benefits from bidirectional context for Arabic MDD [1].
SSL encoder (wav2vec, HuBERT, Whisper)	Rich contextual representations; strong phoneme recognition; robust to speaker and recording variation; effective under limited labelled data.	Computationally expensive; requires large pre-trained models.	Fine-tuned wav2vec achieves 3.2% PER, outperforming human raters on Arabic [6]; Transformer models show competitive results on Arabic phoneme MDD [8].
Multimodal fusion	Combines local and global representations; improves discrimination when evidence is complementary across feature scales.	Architecture complexity; requires parallel branches.	VSCF fusion of MFCC and ResNet50 outperforms single-branch CNN by 1+ percentage points on Qur'anic audio [12].

The pattern across Table 1.3 is consistent: no single architecture is universally optimal. CNNs handle local patterns well, recurrent models capture temporal structure, and SSL encoders excel at content recognition and alignment. This supports a modular design strategy in which each rule category is analysed by the architecture best matched to its dominant acoustic evidence.

1.3.7 Research Gap and Positioning of the Present Work

The review above reveals a consistent pattern across the literature. Existing systems make genuine progress on individual aspects of the problem detecting specific rules, recognising recited content, or identifying pronunciation errors. But none of them brings these aspects together into a single, coherent assessment system that also tells the learner what went wrong and how to fix it.

Three specific gaps stand out.

Limited coverage: Most Tajweed-specific systems handle only a small number of rules and return a simple correct-or-incorrect verdict. A learner who receives such a verdict knows that something was wrong but has no way of knowing where in the recitation the problem occurred or what caused it.

No separation of error types: None of the reviewed systems distinguishes explicitly between content-level errors (Lahn Jali) and rule-level errors (Lahn Khafi) in its output. This distinction matters enormously in practice: telling a learner that they mispronounced a letter requires entirely different guid-

ance from telling them that their Madd was too short.

No rule-aware feedback: No reviewed system connects its detection output to an annotated Qur'anic reference so that it can name the violated rule and indicate precisely where in the text the error occurred. Without this link, a system cannot produce the kind of targeted, position-specific feedback that a qualified teacher would give.

Two broader observations reinforce these conclusions. Birgun's survey of deployed applications [7] confirms that the gap between detecting errors and explaining them is not limited to a few isolated systems it is a field-wide pattern.

This thesis addresses all three gaps. It treats Tajweed assessment as a modular, reference-aware diagnostic task: the system first verifies that the correct content was recited, then checks each applicable rule at each position, and finally uses an annotated Qur'anic reference to produce named, location-specific corrective feedback.

1.3.8 Conclusion

This chapter established the background and context for the work presented in the rest of the thesis. The first part introduced the oral tradition of Qur'anic recitation, the system of canonical transmission variants, the main Tajweed rule families and their acoustic characteristics, the classical distinction between content-level and rule-level errors, and the pedagogical role of the teacher and the coloured Mushaf.

The second part surveyed the state of the art. The field has progressed from early rule-classification systems through sequence-based temporal models to the current generation of large pre-trained speech models. Despite this progress, no existing system combines content verification, rule-aware diagnosis, and corrective feedback into a single pipeline. This gap directly motivates the architecture developed in the following chapter.

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